

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 12: Surface Areas and Volumes

Previous year board questions, topic-wise: 7 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Surface area

1. Find the total surface area of a solid hemisphere whose diameter is $2d$. (a) $3\pi d^2$ (b) $2\pi d^2$ (c) $\frac{1}{2}\pi d^2$ (d) $\frac{3}{4}\pi d^2$ **CBSE 2026, 1 mark**
2. Assertion (A): Two identical solid hemispheres joined along their flat faces form a sphere. Reason (R): The total surface area of a sphere of radius r is $3\pi r^2$. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true **CBSE 2025, 1 mark**
3. A solid sphere is cut into two hemispheres. Find the ratio of the surface area of the sphere to the total surface area of the two hemispheres together. (a) 1 : 1 (b) 1 : 4 (c) 2 : 3 (d) 3 : 2 **CBSE 2024, 1 mark**

2 Volume

4. A solid is a cylinder with a hemisphere on each end. The solid is 20 cm long overall and the cylinder is 7 cm in diameter. Find the volume of the solid. Take $\pi = \frac{22}{7}$. **CBSE 2026, 3 marks**
5. A room has the shape of a cylinder with a hemispherical dome on top. The radius of the dome is half the height of the cylindrical part, and the room holds $\frac{1408}{21}$ m³ of air. Find the height of the cylindrical part. Take $\pi = \frac{22}{7}$. **CBSE 2025, 3 marks**
6. The largest possible right circular cone is cut from a solid cube of edge 2 cm. Find its volume. (a) $\frac{4\pi}{3}$ cu cm (b) $\frac{5\pi}{3}$ cu cm (c) $\frac{8\pi}{3}$ cu cm (d) $\frac{2\pi}{3}$ cu cm **CBSE 2024, 1 mark**
7. A hollow cylinder is 14 cm long. Its outer radius is 1 cm more than its inner radius, and the metal in it has volume 176 cm³. Find the two radii. **CBSE 2024, 3 marks**

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.